JC4004 – Computational Intelligence

Sessions 5 & 6: Game Theory - An introduction of concepts

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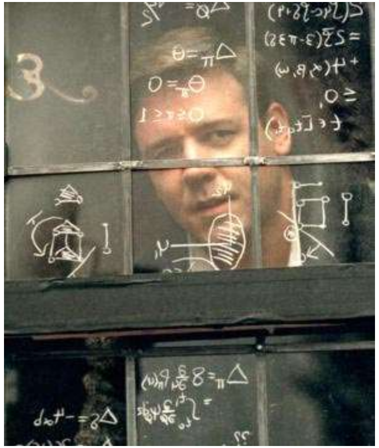
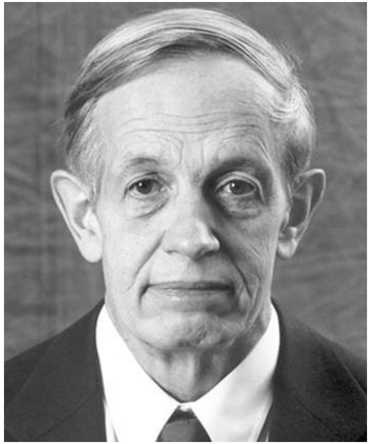
“A Beautiful Mind” (2001)

- Hollywood biopic about mathematician John F. Nash Jr., Nobel laureate in Economics (1994).
- The film dramatizes Nash’s struggles with mental illness and his groundbreaking work in game theory.
- Nash developed the concept of Nash equilibrium, a cornerstone of modern economics, political science, and computer science.
- Popular scenes (e.g. the “bar scene”) illustrate the idea that individuals’ choices depend on anticipating others’ strategies.
- This story connects theory with real life: strategic thinking influences competition, cooperation, and decision-making in society.





pic source: https://yung-web.github.io/home/courses/gametheory.html

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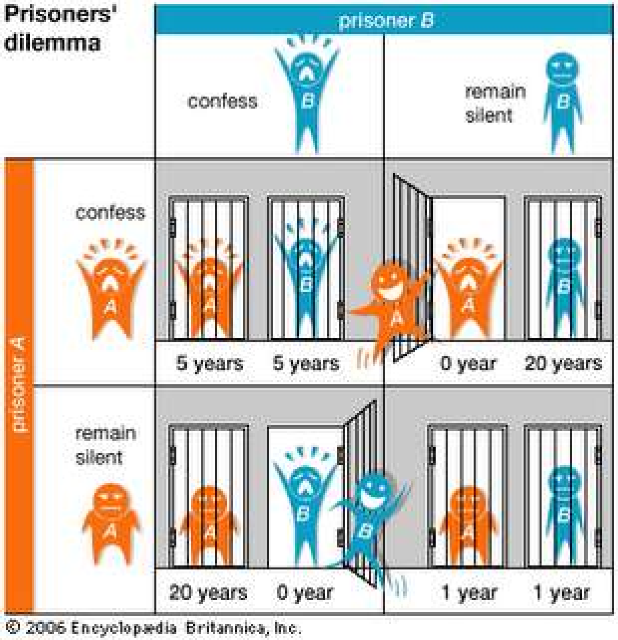
Prisoner’s Dilemma

- *Two suspects, A and B, are arrested and interrogated separately*.
- Each has two choices: Confess or Remain Silent.
- Payoffs (years in prison) depend on the combination of their decisions:
  - If both confess → both get 5 years.
  - If A confesses but B stays silent → A goes free, B gets 20 years (and vice versa).
  - If both stay silent → both get 1 year (for a minor charge).
- **Dilemma**:
  - Mutual cooperation (both silent) is better for them jointly (1+1 = 2 years total).
  - But the dominant strategy is to confess, leading to (5,5) — a worse outcome for both.
  - This illustrates the tension between individual rationality and collective welfare, a central theme in game theory.



pic source: https://yung-web.github.io/home/courses/gametheory.html

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Prisoner’s Dilemma – Payoff Matrix

- Two players: Prisoner A (row) and Prisoner B (column).
- Each chooses: Confess or Remain Silent.
- Payoffs shown as (A’s years, B’s years).
- Lower numbers are better (fewer years in prison).
- Confessing is the dominant strategy → Nash Equilibrium: (Confess, Confess).
- Outcome: (5, 5) – worse for both compared to cooperation (1, 1).

	B: Confess	B: Silent
A: Confess	5, 5	0, 20
A: Silent	20, 0	1, 1

The “2/3 of the average” game

Everyone writes down a number between 0 and 100, the one closest to 2/3 of the average wins.

Example:

- A says 50
- B says 10
- C says 90
- Average(50, 10, 90) = 50
- 2/3 of average = 33.33

A is closest (|50-33.33| = 16.67), so A wins

Player	Choice	Distance to 2/3 Avg
A	50	|50 - 33.3| = 16.7
B	10	|10 - 33.3| = 23.3
C	90	|90 - 33.3| = 56.7

Example source: Duke University “Game Theory” (https://courses.cs.duke.edu/spring22/compsci323d/cm_game_theory.ppt)

The “2/3 of the average” game

Rational analysis:

- If everyone reasons one step: they avoid high numbers (closer to 2/3).

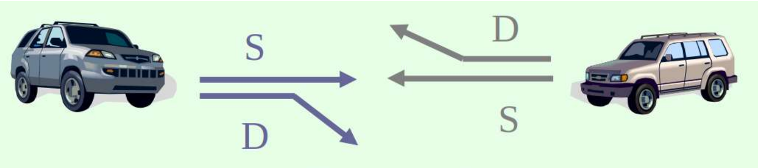
- Iterated reasoning drives choices downward: 2/3 of 100 = 66.7, then 44.4, 29.6, etc.

- The unique Nash Equilibrium: all players choose 0.

Example source: Duke University “Game Theory” (https://courses.cs.duke.edu/spring22/compsci323d/cm_game_theory.ppt)

The “Chicken” game

- Two players drive cars towards each other.
- If one player goes straight, that player wins; If both go straight, they both die.



	B: D (Dodge)	B: S (Straight)
A: D	0, 0	-1, 1
A: S	1, -1	-5, -5

Example source: Duke University “Game Theory” (https://courses.cs.duke.edu/spring22/compsci323d/cm_game_theory.ppt)

The "Chicken" game

Two players drive towards each other. Choices:

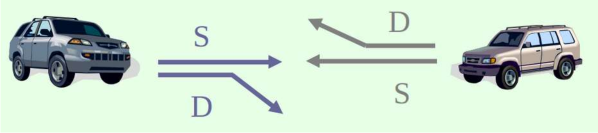
- D (Dodge/Swerving): avoid collision.
- S (Straight): keep going straight.

Payoffs (Player A, Player B):

- (S, S): Both go straight → catastrophic crash (–5, –5).
- (D, D): Both dodge → neutral outcome (0, 0).
- (S, D): Straight vs. Dodge → Straight 'wins' (1), Dodge 'loses' (–1).

Analysis:

- Two pure-strategy Nash equilibria: (S, D) and (D, S).
- Each player prefers to go straight if the other dodges.
- There is also a mixed-strategy equilibrium (randomizing S and D).
- Game captures conflict between aggression and caution.



	B: D (Dodge)	B: S (Straight)
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A: S	1, -1	-5, -5

Example source: Duke University "Game Theory" (https://courses.cs.duke.edu/spring22/compsci323d/cm_game_theory.ppt)

‘Rock-paper-scissors’

- Player 1 (row) and Player 2 (column) play the same simultaneously.
- The table on the right lists payer 1 & 2 rewards (utility)
- Row player’s utility is always listed first, column player’s second
- Zero-sum game: the utilities in each entry sum to 0 (or a constant)
- Three-player game would be a 3D table with 3 utilities per entry, etc.

	Rock (P2)	Paper (P2)	Scissors (P2)
Rock (P1)	0, 0	-1, 1	1, -1
Paper (P1)	1, -1	0, 0	-1, 1
Scissors (P1)	-1, 1	1, -1	0, 0

What is Game Theory?

- Game theory is the mathematical study of strategic interaction.
- It analyzes how decision-makers (players) choose actions that affect each other’s outcomes.
- Applications span economics, political science, computer science, biology, and everyday life.
- Focuses on rational decision-making under interdependence.

Key Elements of a Game

1. Players: decision-makers in the game.
2. Strategies: available actions (pure or mixed).
3. Payoffs: outcomes (utilities, profits, costs).
4. Information: what players know when making choices.
5. Equilibrium: stable outcome where no player wants to deviate.

Types of Games

- Cooperative *vs.* Non-cooperative games
- Zero-sum *vs.* Non-zero-sum games
- Simultaneous *vs.* Sequential games
- Perfect *vs.* Imperfect information
- Finite *vs.* Infinite horizon games

Applications of Game Theory

- Economics: market competition, auctions, bargaining.
- Politics: voting systems, coalition formation, deterrence.
- Computer Science: algorithms, cybersecurity, AI decision-making.
- Biology: evolution, cooperation, animal behavior.
- Everyday Life: negotiations, traffic, social dilemmas.

Strategic-Form Games

- A strategic-form (normal-form) game is defined by:

  - A set of players N.

  - For each player $i \in N$, a set of strategies S_i.

  - For each player $i \in N$, a payoff function $u_i: S \to \mathbb{R}$.

Payoff Matrix

- Players: The decision-makers in the game (e.g., Player A and Player B).
- Strategies: The available actions for each player (e.g., A chooses Up/Down, B chooses Left/Right).
- Matrix cells: Each cell corresponds to a combination of strategies, one from each player.
- Payoffs: The ordered pair inside each cell shows the outcome for both players.
  - The first number is the payoff to the row player (Player A).
  - The second number is the payoff to the column player (Player B).

The matrix summarizes the game in strategic (normal) form, showing what each player receives depending on their joint actions.

	Player B: L	Player B: R
Player A: U	(2,2)	(0,3)
Player A: D	(3,0)	(1,1)

Strict and Weak Dominance

- Strategy s_i strictly dominates s'_i if it always yields a higher payoff.
- Weak dominance: at least as good in all cases, strictly better in some.
- Rational players never play strictly dominated strategies.

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Example: Strict Dominance

	B: Left	B: Right
A: Up	(2,1)	(1,0)
A: Down	(3,1)	(2,0)

If B plays Left → A gets 2 vs 3.
If B plays Right → A gets 1 vs 2.
Down strictly dominates Up (always better).

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Example: Weak Dominance

	B: Left	B: Right
A: Up	(2,1)	(1,0)
A: Down	(2,1)	(1,1)

If B plays Left → both give 2.
If B plays Right → both give 1.
Down weakly dominates Up (never worse, sometimes equal).

Iterated Deletion of Dominated Strategies

- Eliminate dominated strategies step by step to simplify the game.
- IDSDS (Strict Dominance):
    - A strategy is strictly dominated if another always gives a higher payoff.
    - Rational players never play strictly dominated strategies.
    - Delete strictly dominated strategies iteratively until none remain.
    - Result is unique (independent of elimination order).

Iterated Deletion of Dominated Strategies

- IDWDS (Weak Dominance):
  - A strategy is weakly dominated if another is never worse and strictly better in some cases.
  - Rational players might still use weakly dominated strategies.
  - Iterated deletion possible, but results may depend on elimination order.
  - Less robust than IDSDS.
- IDSDS is a reliable tool to shrink games to their 'strategic core,' while IDWDS can be ambiguous.

Strategic-Form Games: Summary

- Iterated deletion simplifies games
- Strict dominance guarantees unique results
- Weak dominance can lead to multiple outcomes
- IDSDS effectively reduces game complexity

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Rationalizability

- A strategy is rationalizable if it is a best response to some belief about others' strategies.
- It generalizes iterated dominance: instead of eliminating only dominated strategies, we keep strategies that could be justified as optimal under some belief.
- Example:

For Player A:

If A believes B will play L with high probability, U is a best response.

If A believes B will play R with high probability, D is a best response.

Therefore, both U and D are rationalizable (each can be justified by some belief).

	B: L	B: R
A: U	2,2	0,3
A: D	1,1	1,0

Nash Equilibrium

- A **strategy profile** $s^* = (s_1^*, \ldots, s_n^*)$ is a Nash Equilibrium if no player can improve their payoff by deviating *unilaterally*.
- Formally:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall\, s_i \in S_i,\ \forall i \in N$$

  - s_i^*: player i's chosen equilibrium strategy.
  - s_{-i}^*: the strategies chosen by all other players.
  - u_i: payoff function of player i.

- At equilibrium, each player's strategy is a best response to the others.
- No player has an incentive to deviate, given the others' choices.
- Stability: once reached, the outcome tends to persist.

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Nash Equilibrium

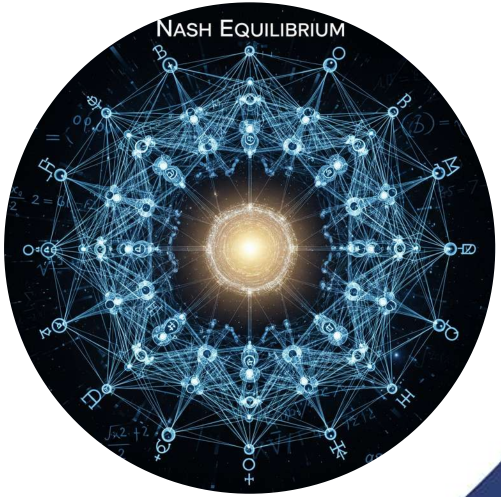
- Equilibrium ≠ optimal for all players collectively (e.g., Prisoner’s Dilemma).
- NE can be in **pure strategies** (deterministic actions) or **mixed strategies** (randomization).
- Every finite game has at least one Nash Equilibrium (possibly in mixed strategies).

Existence of Nash Equilibrium

Nash (1950): Every finite game has at least one mixed-strategy NE.

Proof uses fixed-point theorem (Kakutani).

This ensures predictive power of game theory.



Pure vs Mixed Equilibria

- Pure equilibrium: deterministic choice.
- Mixed equilibrium: probability distribution over pure strategies.
- Example: Rock–Paper–Scissors has no pure NE, only mixed NE.

Pure Strategy Equilibrium

A pure strategy means a player always makes the same deterministic choice.

- Example: *Prisoner’s Dilemma*
  - If both players choose 'Confess', neither can gain by switching to 'Silent'.
  - (Confess, Confess) is a Pure Strategy Nash Equilibrium.

- Pure strategies contrast with mixed strategies, where players randomize.

Mixed Strategies

- A mixed strategy for player i is a probability distribution over S_i.
- Expected utility is computed by averaging payoffs over probability distributions.

e.g. the **Rock–Paper–Scissors game**:

In this game, no **pure strategy** (always playing Rock, always Paper, or always Scissors) is stable because each pure strategy can be beaten.

- The **mixed strategy Nash equilibrium** is when each player randomizes uniformly:
  - Rock with probability 1/3
  - Paper with probability 1/3
  - Scissors with probability 1/3.

This way, opponents cannot predict or exploit a pattern. The expected payoff for each player is 0, meaning the game is fair in equilibrium.

Multiple Equilibria

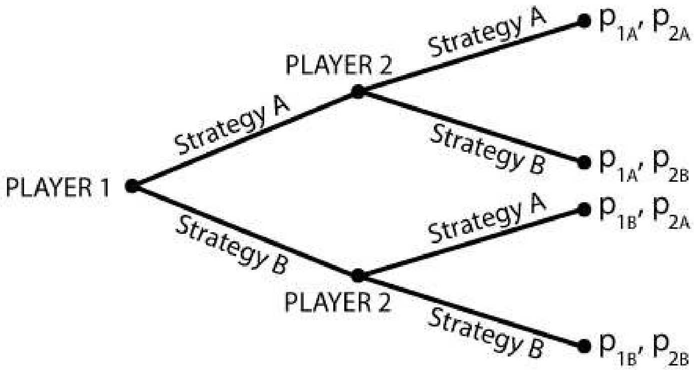
- Some games have multiple Nash equilibria.
- Coordination games: more than one stable outcome.
- Selection depends on conventions, history, focal points.

Zero-Sum vs Non-Zero-Sum

- Zero-sum: $u_A + u_B$ = constant (e.g., Matching Pennies).
- Non-zero-sum: total payoffs vary (e.g., Prisoner's Dilemma).
- Different solution approaches: minimax vs NE.

Extensive Form: Game Trees

- Represents sequential structure of moves.
- Nodes = decision points, edges = actions, leaves = outcomes.
- Information sets capture imperfect information.



Perfect Information Games

- Every player knows all prior moves.
- Chess, checkers are classic examples.
- Formalized as finite game trees.

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Backward Induction

An iterative process of reasoning backward from the end of a problem or situation to solve finite extensive-form and sequential games and infer a sequence of optimal actions.

- Solution for finite perfect-information games.
- Start at terminal nodes, move backward selecting optimal actions.
- Yields subgame-perfect equilibrium.

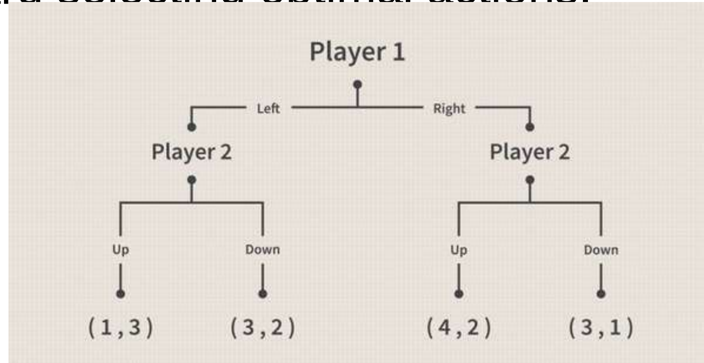

Figure source: INvestopedia: https://www.investopedia.com/terms/b/backward-induction.asp

Backward Induction vs Nash Equilibrium

- Every backward induction outcome is a Nash equilibrium.
- But not every Nash equilibrium satisfies backward induction.
- Bonanno Thm 3.4.1 clarifies the relation:

> **Theorem 3.4.1** Every backward-induction solution of a perfect-information game is a Nash equilibrium of the associated strategic form.

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Ref: Game Theory, pp.87,
https://faculty.econ.ucdavis.edu/faculty/bonanno/PDF/GT_book.pdf

Subgames

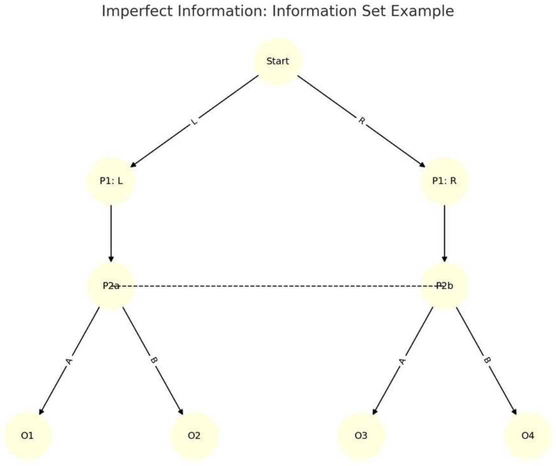
- A subgame starts at a decision node and contains all successors.
- Subgames must respect information sets.
- Used to define subgame-perfect equilibrium.

Subgame-Perfect Equilibrium

- A strategy profile is SPE if it induces a NE in every subgame.
- Eliminates non-credible threats.
- Refines Nash equilibrium.

Imperfect Information

- Players may not observe others’ choices.
- Represented by information sets (indistinguishable nodes).
- Complicates equilibrium analysis.



Player 1 moves first (chooses L or R).

Player 2 then moves, but cannot distinguish whether Player 1 chose L or R → represented by the **dashed line** (an information set).

Outcomes (O1–O4) result depending on Player 2’s choice (A or B).

Chance Moves

- Some games include random events ('Nature').
- Nature chooses according to fixed probability distribution.
- Example: card games with shuffled decks.

Expected Utility Theory

- Von Neumann-Morgenstern framework.
- Players evaluate lotteries by expected utility.
- Assumes rationality axioms (completeness, transitivity, independence).

Von Neumann–Morgenstern Utility Theorem

If a player's preferences over risky choices (lotteries) satisfy certain rationality axioms, then there exists a utility function u such that the player behaves as if they are maximizing expected utility.

The theorem guarantees that rational choice under uncertainty can be modeled as maximizing expected utility.

Von Neumann–Morgenstern Utility Theorem

Axioms required (preferences must satisfy):

1. **Completeness** – For any two lotteries A and B, the player can say either $A \succeq B$, $B \succeq A$, or is indifferent.
2. **Transitivity** – If $A \succeq B$ and $B \succeq C$, then $A \succeq C$.
3. **Continuity** – If $A \succeq B \succeq C$, then there is some probability p such that B is indifferent to a lottery that gives A with probability p and C with probability $1 - p$.
4. **Independence** – If $A \succeq B$, then for any C and any probability p, the player prefers $pA + (1 - p)C \succeq pB + (1 - p)C$.

Implication:

- Preferences can be represented by a **cardinal utility function** u.
- Players evaluate a lottery $L = \{p_1, x_1; p_2, x_2; \ldots; p_n, x_n\}$ by computing

$$EU(L) = \sum_{i=1}^{n} p_i \, u(x_i).$$

- The player chooses the lottery with the highest expected utility.

Common Knowledge

An event is common knowledge if:

- Everyone knows it,
- Everyone knows that everyone knows it, and so on ad infinitum.

A key role in coordination and equilibrium selection.

Rationality & Common Knowledge

- Rationality: Players choose strategies that maximize their payoffs.
- Common knowledge of rationality: Everyone knows all players are rational, and knows that everyone knows it, ad infinitum.
- Implications:
  - Strictly dominated strategies are eliminated.
  - Iterated elimination (IDSDS/IDWDS) reduces the game to a strategic core.
- Rationalizability: Strategies that survive common knowledge of rationality. Broader than Nash equilibrium.
- e.g. In the Prisoner’s Dilemma, 'Confess' is rationalizable since 'Silent' is strictly dominated.

Bayesian Games

- Players have private information ('types').
- Beliefs over types represented by probability distributions.
- Strategies map types into actions.

Bayesian Nash Equilibrium

- A strategy profile where each type’s strategy maximizes expected payoff given beliefs.
- Generalizes Nash equilibrium to incomplete information.

Evolutionary Stability

- Introduced in evolutionary game theory (Maynard Smith & Price, 1973).
- Extends Nash equilibrium to evolutionary contexts where strategies are inherited and tested by natural selection.

A strategy s^* is an **evolutionarily stable strategy (ESS)** if, when most of the population plays s^*, no rare mutant strategy s' can do better.

- Formally:
  1. $u(s^*, s^*) \geq u(s', s^*)$ for all s'.
  2. If $u(s^*, s^*) = u(s', s^*)$, then $u(s^*, s') > u(s', s')$.

Intuition:

Once an ESS takes over, it is **resistant to invasion** by alternative strategies.

It is a refinement of Nash equilibrium that incorporates **stability under perturbations**.

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Equilibrium Concepts: Summary

Concept	Definition	Context
Nash Equilibrium	No unilateral deviation improves payoff	All games
Subgame-Perfect Equilibrium	NE in every subgame	Dynamic games
Bayesian Nash Equilibrium	NE with private types & beliefs	Incomplete info
ESS	Stable against invasion	Evolutionary games

Types of Games

Dimension	Options
Cooperation	Cooperative vs Non-cooperative
Payoffs	Zero-sum vs Non-zero-sum
Timing	Simultaneous vs Sequential
Information	Perfect vs Imperfect
Completeness	Complete vs Incomplete

Summary

- Strategic-form games: dominance, rationalizability, Nash equilibrium.
- Extensive-form games: backward induction, subgame-perfect equilibrium.
- Uncertainty: expected utility, Bayesian Nash equilibrium.
- Advanced: common knowledge, evolutionary stability.

Textbooks / References

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- P.K. Dutta. <*Strategies and Games: Theory And Practice*>, MIT 1999

Questions and discussion

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